Chapter 5 Exercises: Selected Answers and Solutions

1. If X is the diameter of a shaft, we are told that

$$X \sim N\left(1.005, 0.01^2\right).$$

The specification set is $S = (0.98, 1.02)$ and so

$$\begin{aligned} P(X \in S) &= P(0.98 < X < 1.02) \\ &= P\left(\frac{0.98 - 1.005}{0.01} < Z < \frac{1.02 - 1.005}{0.1}\right) \\ &= P(-2.5 < Z < 1.5) \\ &= P(Z < 1.5) - P(Z < -2.5) \\ &= 0.9332 - 0.0062 \\ &= 0.927. \end{aligned}$$

So $P(X \notin S) = 1 - 0.927 = 0.073$, hence about 7.3% will fall outside the specification limits. Obviously setting the mean of X to be the centre of S maximises the chance of attaining the specification, so $\mu = 1$ is optimal.

2. xxx

3. Let $\Phi(\cdot)$ and $\phi(\cdot)$ denote the c.d.f and p.d.f of the standard Normal distribution respectively. If $X_1, X_2 \sim N(0, 4)$ are independent, then let $Y = \max\{X_1, X_2\}$ have c.d.f. $F(y)$. Hence

$$\begin{aligned} F(y) &= P(Y < y) \\ &= P(X_1 < y, X_2 < y) \\ &= \Phi\left(\frac{y}{2}\right)\Phi\left(\frac{y}{2}\right) \\ &= \Phi\left(\frac{y}{2}\right)^2, \end{aligned}$$

by independence and the fact that the c.d.f. of X_i is $\Phi\left(\frac{y}{2}\right)$ for $i = 1, 2$. The p.d.f. of Y is

$$f(y) = \frac{dF(y)}{dy}$$

$$
\begin{aligned}
&= \frac{d}{dy}\Phi\left(\frac{y}{2}\right)^2 \\
&= 2\Phi\left(\frac{y}{2}\right)\frac{d}{dy}\Phi\left(\frac{y}{2}\right).
\end{aligned}
$$

Putting $u = y/2$,

$$
\frac{du}{dy} = \frac{1}{2}
$$

and so

$$
\begin{aligned}
\frac{d}{dy}\Phi\left(\frac{y}{2}\right) &= \frac{d\Phi(u)}{du} \times \frac{du}{dy} \\
&= \phi(u) \times \frac{1}{2} \\
&= \frac{\phi(y/2)}{2}.
\end{aligned}
$$

Hence

$$
\begin{aligned}
f(y) &= 2\Phi\left(\frac{y}{2}\right)\frac{\phi(y/2)}{2} \\
&= \Phi\left(\frac{y}{2}\right)\phi\left(\frac{y}{2}\right).
\end{aligned}
$$

4. xxx

5. xxx

6. xxx

7. If X is the length of a board, we know $X \sim N(\mu, 0.2^2)$ for some μ. Now we require

$$
\begin{aligned}
P(X < 8) &= P\left(\frac{X-\mu}{0.2} < \frac{8-\mu}{0.2}\right) \\
&= P\left(Z < \frac{8-\mu}{0.2}\right) \\
&= 0.05.
\end{aligned}
$$

From the tables, $P(Z < z) = 0.05$ when $z = -1.645$, hence

$$
\frac{8-\mu}{0.2} = -1.645
$$

which implies that μ is 8.329 feet.

8. xxx

9. .

   (a) 0.0456

   (b) $E(X^2) = 2525$, $IQR = 6.75$.

10. .

    (a) 0.9032

    (c) 0.6913

    (e) $c = 1.3$

    (g) $c = 1.645$

    (i) $c = 0.255$

11. With $X \sim N(10, 25)$ let

$$Z = \frac{X - 10}{5} \sim N(0, 1)$$

as usual.

(a)

$$\begin{aligned} P(X < 13) &= P\left(\frac{X - 10}{5} < \frac{13 - 10}{5}\right) \\ &= P(Z < 0.6) \\ &= 0.7257. \end{aligned}$$

(b)

$$\begin{aligned} P(11 < X < 13) &= P\left(\frac{11 - 10}{5} < \frac{X - 10}{5} < \frac{13 - 10}{5}\right) \\ &= P(0.2 < Z < 0.6) \\ &= P(Z < 0.6) - P(Z < 0.2) \\ &= 0.7257 - 0.5793 \\ &= 0.1464. \end{aligned}$$

(c)

$$\begin{aligned} P(8 < X < 13) &= P\left(\frac{8-10}{5} < \frac{X-10}{5} < \frac{13-10}{5}\right) \\ &= P(-0.4 < Z < 0.6) \\ &= P(Z < 0.6) - P(Z < -0.4) \\ &= P(Z < 0.6) - (1 - P(Z < 0.4)) \\ &= 0.7257 - 1 + 0.6554 \\ &= 0.3811. \end{aligned}$$

(d)

$$\begin{aligned} P(6 < X < 8) &= P\left(\frac{6-10}{5} < \frac{X-10}{5} < \frac{8-10}{5}\right) \\ &= P(-0.8 < Z < -0.4) \\ &= P(Z < -0.4) - P(Z < -0.8) \\ &= 0.3446 - 0.2118 \\ &= 0.1327. \end{aligned}$$

(e) We require c such that

$$P(X < c) = 0.9032,$$

equivalently,

$$P\left(\frac{X-10}{5} < \frac{c-10}{5}\right) = 0.9032$$

or

$$P\left(Z < \frac{c-10}{5}\right) = 0.9032.$$

This implies

$$\frac{c-10}{5} = 1.3,$$

and so $c = 16.5$.

(f) To find c such that

$$P(X < c) = 0.0968,$$

we require

$$P\left(\frac{X-10}{5} < \frac{c-10}{5}\right) = 0.0968.$$

Equivalent to this is the statement

$$\begin{aligned} P\left(Z < \frac{c-10}{5}\right) &= 0.0968 \\ &= 1 - 0.9032. \end{aligned}$$

This implies

$$\frac{c-10}{5} = -1.3,$$

and so $c = 3.5$.

(g) Now if

$$P\left(-c < X - 10 < c\right) = 0.9$$

then

$$P\left(-\frac{c}{5} < \frac{X-10}{5} < \frac{c}{5}\right) = 0.9.$$

Hence

$$\begin{aligned} P\left(-\frac{c}{5} < Z < \frac{c}{5}\right) &= P\left(Z < \frac{c}{5}\right) - P\left(Z < -\frac{c}{5}\right) \\ &= P\left(Z < \frac{c}{5}\right) - \left(1 - P\left(Z < \frac{c}{5}\right)\right) \\ &= 2P\left(Z < \frac{c}{5}\right) - 1 \\ &= 0.9. \end{aligned}$$

Therefore

$$P\left(Z < \frac{c}{5}\right) = 0.9$$

and hence

$$c = 5 \times 1.645 = 8.225.$$

(h) xxx

(j) When

$$P\left(|X - 10| > c\right) = 0.8$$

we have

$$\begin{aligned} P\left(|Z| > \frac{c}{5}\right) &= 0.8 \\ &= 1 - P\left(|Z| < \frac{c}{5}\right) \\ &= 1 - \left(2P\left(Z < \frac{c}{5}\right) - 1\right) \end{aligned}$$

and so

$$P\left(Z < \frac{c}{5}\right) = 0.6$$

and hence

$$\frac{c}{5} \approx 0.26$$

and so $c \approx 1.3$.

(h)

$$\begin{aligned} P\left(-c < X < c\right) &= P\left(X < c\right) - P\left(X < -c\right) \\ &= P\left(Z < \frac{c-10}{5}\right) - P\left(Z < -\left(\frac{c-10}{5}\right)\right) \\ &= P\left(Z < \frac{c-10}{5}\right) - \left(1 - P\left(Z > -\left(\frac{c-10}{5}\right)\right)\right) \\ &= 2P\left(Z < \frac{c-10}{5}\right) - 1 \\ &= 0.95. \end{aligned}$$

Hence

$$P\left(Z < \frac{c-10}{5}\right) = 0.975$$

and so

$$\begin{aligned} c &= 5 \times 1.96 + 10 \\ &= 19.8. \end{aligned}$$

12. 3.9

13. .

    (a) 0.8664

(b) 0.516.

14. .

(a) 0.805.

(b) $\sigma = 1.8175$.

15. Let X_i denote the lifetime in hours of an electron tube, for $i = 1, 2, 3, 4$. Then

$$X_i \sim N\left(95, 6^2\right),$$

and we will assume that these four variables are independent (though this is not explicitly stated in the question).

(a) Hence

$$\begin{aligned} P(X > 100) &= 1 - P(X < 100) \\ &= 1 - P\left(\frac{X - 95}{6} < \frac{100 - 95}{6}\right) \\ &= 1 - P(Z < 0.833) \\ &= 1 - 0.7975 \\ &= 0.2025. \end{aligned}$$

(b) If Y denotes the lifetime of the system, then

$$\begin{aligned} P(Y > 90) &= P(X_1 > 90, X_2 > 90, X_3 > 90, X_4 > 90) \\ &= \prod_{i=1}^{4} P(X_i > 90) \\ &= \prod_{i=1}^{4} (1 - P(X_i < 90)) \\ &= (1 - P(Z < -0.833))^4 \\ &= (1 - 0.2025)^4 \\ &= 0.4045. \end{aligned}$$

16. Let the weights of the components be X_1, X_2 and X_3 where

$$\begin{aligned} X_1 &\sim N(2, 0.02), \\ X_2 &\sim N(1, 0.01), \\ X_3 &\sim N(3, 0.03) \end{aligned}$$

and we assume these variables are independent. Changing notation from the question, let

$$Y = X_1 + X_2 + X_3$$

then

$$Y \sim N\left(2 + 1 + 3, 0.02 + 0.01 + 0.03\right) = N\left(6, 0.06\right).$$

Further,

$$\begin{aligned} P\left(5.7 < Y < 6.3\right) &= P\left(\frac{5.7 - 6}{\sqrt{0.06}} < Z < \frac{6.3 - 6}{\sqrt{0.06}}\right) \\ &= P\left(-1.225 < Z < 1.225\right) \\ &= P\left(Z < 1.225\right) - P\left(Z < -1.225\right) \\ &= 0.8897 - 0.1103 \\ &= 0.7794. \end{aligned}$$

17. Let X be the weight (in hundred pounds) of a beam. Then

$$X \sim N\left(31, 0.5^2\right).$$

(a) So

$$\begin{aligned} P\left(30 < X < 32\right) &= P\left(X < 32\right) - P\left(X < 32\right) \\ &= P\left(\frac{X - 31}{0.5} < 2\right) - P\left(\frac{X - 30}{0.5} < -2\right) \\ &= P\left(Z < 2\right) - P\left(Z < -2\right) \\ &= 0.9772 - 0.0228 \\ &= 0.9544. \end{aligned}$$

(b) If $X_i \sim N\left(31, 0.5^2\right)$ for $i = 1, 2, \ldots, 25$ then define

$$\begin{aligned} Y &= X_1 + \cdots + X_{25} \\ &\sim N\left(25 \times 31, 25 \times 0.5^2\right) \\ &= N\left(775, 6.25\right). \end{aligned}$$

We require

$$\begin{aligned} P(Y > 795) &= 1 - P(Y < 795) \\ &= 1 - P\left(Z < \frac{795 - 775}{2.5}\right) \\ &= 1 - P(Z < 8) \\ &\approx 0. \end{aligned}$$

18. Here $\mu = 250513$. For the larger bags, $\mu = 500.513$.

19. xxx

BD